APEXPLANET CYBERSECURITY & ETHICAL HACKING INTERNSHIP

Task 5: Capstone Project & Incident Response - Final Technical Report

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Program: Cybersecurity & Ethical Hacking Internship (Days 49–60)

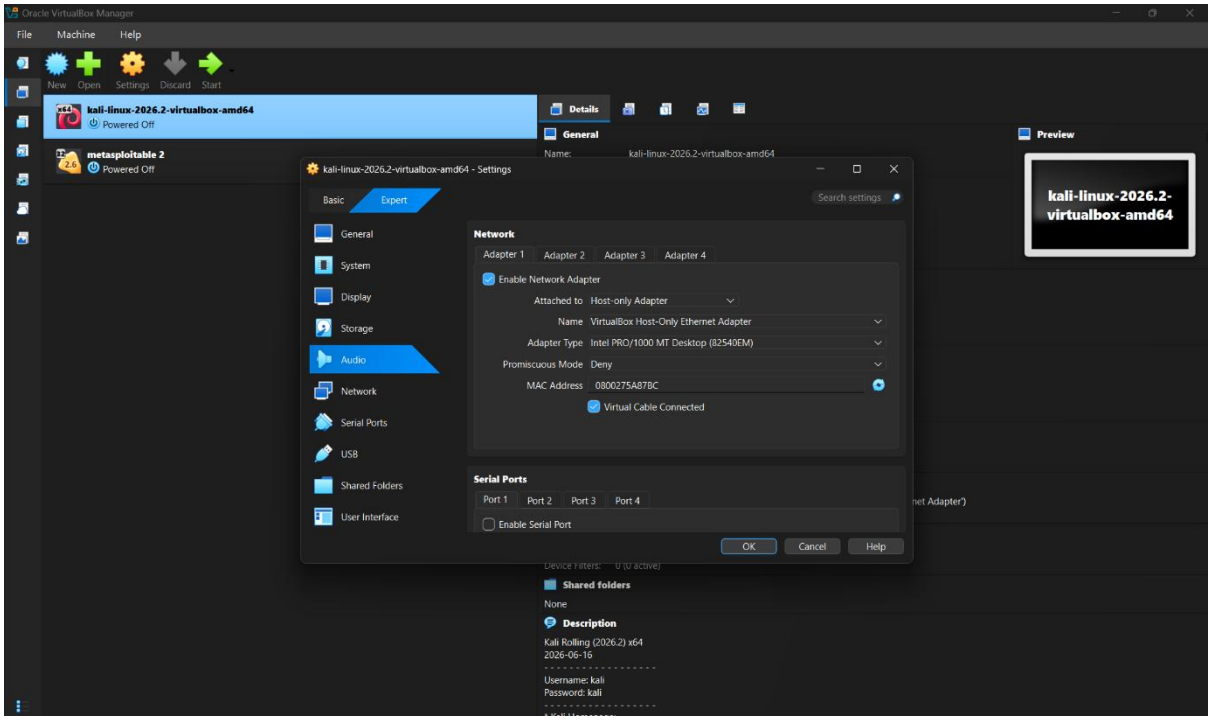
Target Environment IP: 192.168.56.103 (Metasploitable2)

1. Executive Summary

This capstone project report encapsulates the hands-on security assessment, penetration testing workflow, and incident response simulation executed as the final phase of the ApexPlanet internship. The project demonstrates end-to-end competency in setting up an isolated virtual hacking lab, conducting network reconnaissance, identifying vulnerabilities, executing controlled exploitation, and performing incident response (containment, eradication, and recovery).

2. Project Scope & Lab Architecture

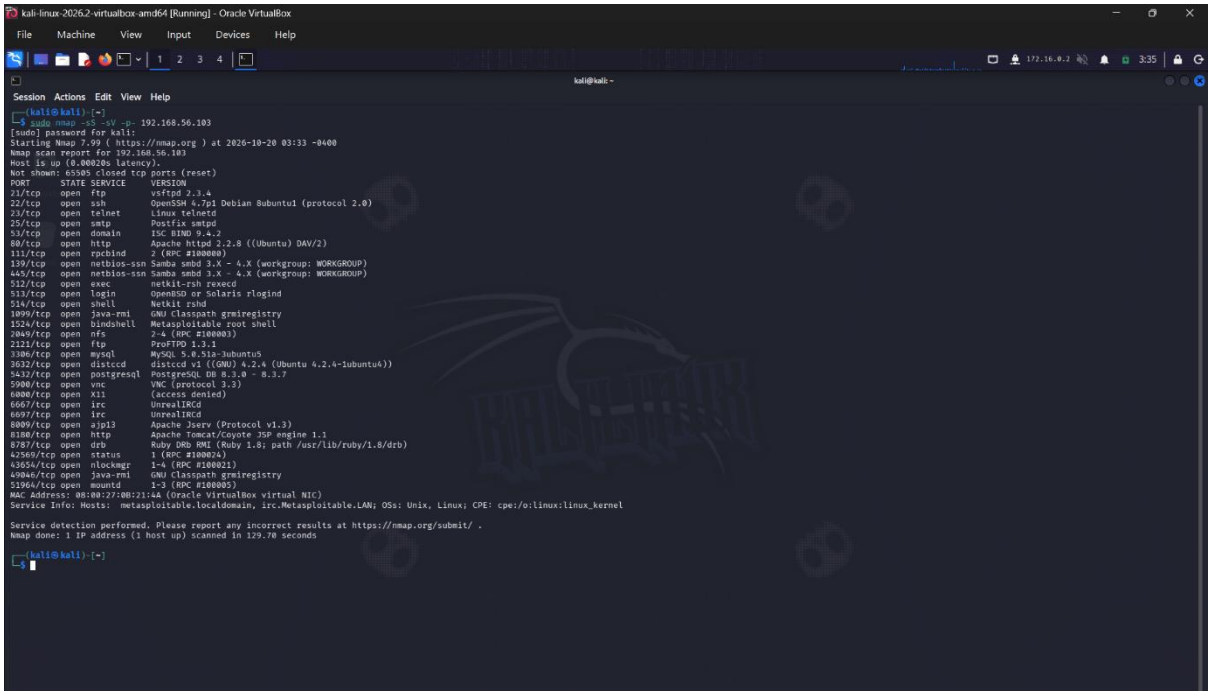
- **Objective:** To assess a vulnerable test network, document vulnerabilities, execute controlled exploitation, and simulate a complete incident response lifecycle.
- **Scope:** Isolated strictly to a local VirtualBox Host-Only network environment to ensure safety and compliance.
- **Lab Topology:**
    - **Attacker Node:** Kali Linux (Network testing platform)
    - **Target Node:** Metasploitable2 (192.168.56.103) (Vulnerable test server)



3. Methodology & Implementation Phases

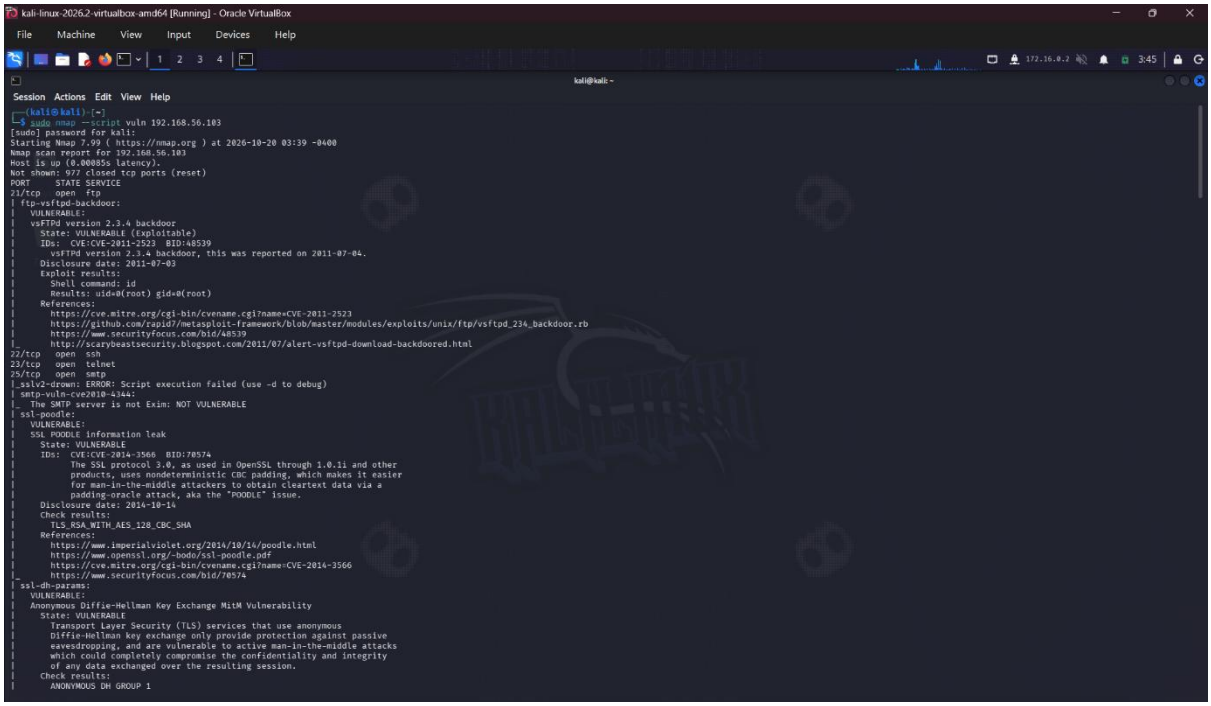
Phase 1: Reconnaissance & Scanning

- **Port & Service Scanning:** Executed Nmap scans against the target node (192.168.56.103) to enumerate open ports, running services, and operating system details.
    - *Command Example:* nmap -sS -sV -O 192.168.56.103
- **Vulnerability Assessment:** Mapped out critical, high, and medium-risk vulnerabilities inherent to the outdated services running on the target machine.



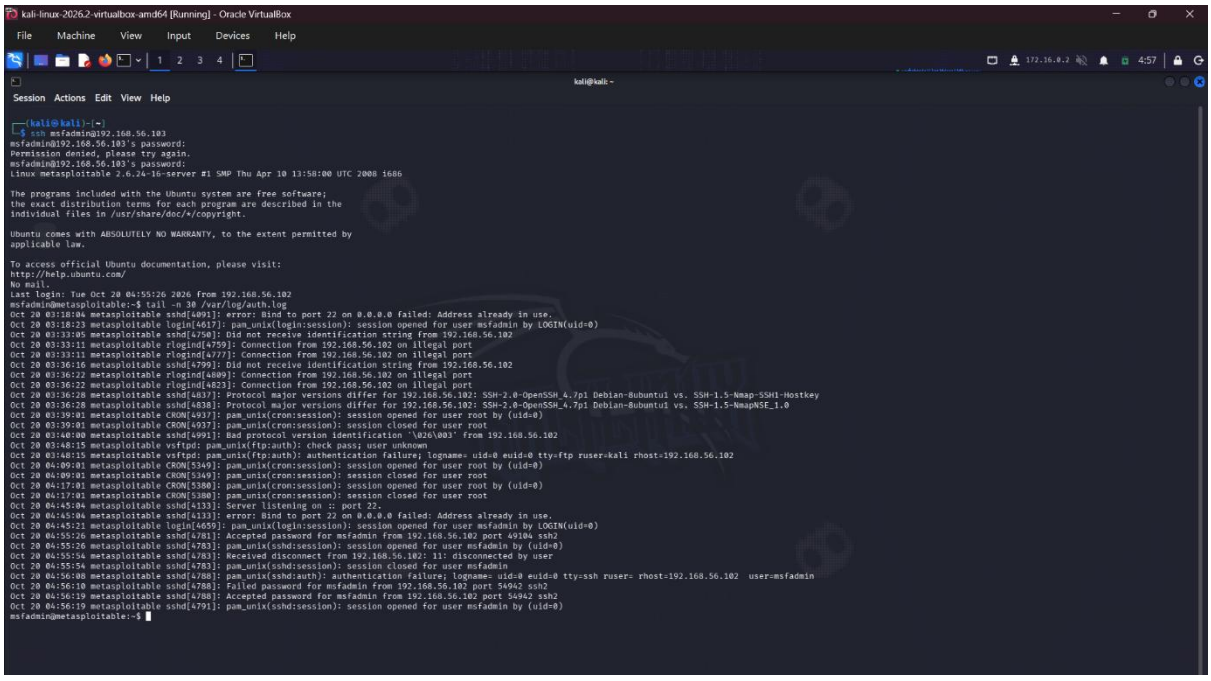
Phase 2: Exploitation

- **Exploitation Workflow:** Leveraged the Metasploit Framework on Kali Linux to target a known vulnerable service on the target machine.
- **Access & Post-Exploitation:** Successfully established a controlled session, verified system information (sysinfo), and demonstrated post-exploitation visibility.



Phase 3: Incident Response Simulation (Detection, Containment & Eradication)

- **Log Detection:** Monitored system and service logs to identify unauthorized connection attempts and anomalous shell activity.
- **Containment:** Immediately disconnected the target interface from the network to halt lateral movement and isolate the threat.
- **Eradication:** Terminated unauthorized active sessions, removed malicious persistence artifacts, and applied service restrictions.



```
msfadmin@metasploitable:~$ sudo killall vsftpd
vsftpd: no process killed
msfadmin@metasploitable:~$ sudo update-rc.d -f inetd remove
 Removing any system startup links for /etc/init.d/inetd ...
msfadmin@metasploitable:~$ sudo invoke-rc.d openbsd-inetd stop
msfadmin@metasploitable:~$ sudo iptables -L -n -v
Chain INPUT (policy ACCEPT 233 packets, 58422 bytes)
 pkts bytes target     prot opt in     out     source               destination

   27  2428 DROP       all  --  *      *       192.168.56.102       0.0.0.0/0

    0     0 DROP       all  --  *      *       192.168.56.102       0.0.0.0/0


Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination


Chain OUTPUT (policy ACCEPT 232 packets, 57587 bytes)
 pkts bytes target     prot opt in     out     source               destination

msfadmin@metasploitable:~$ _
```

4. Findings & Evidence Summary

Vulnerability / Event	Severity	Affected Service / Port	Description & Impact
Unencrypted Services / Weak Daemons	High	FTP / Telnet	Legacy protocols permitting cleartext credential transmission.
Vulnerable Application Backdoors	Critical	Custom / Web / RPC	Outdated service configurations exploitable via Metasploit modules.
Unauthorized Session Activity	Critical	System Shell	Simulated reverse shell execution demonstrating host compromise.

5. Mitigation Strategies

To eliminate the identified vulnerabilities and harden the environment against future attacks, the following controls are recommended:

1. **Service Hardening:** Disable legacy, unencrypted protocols (such as Telnet and unencrypted FTP) and migrate entirely to secure alternatives (SSH, SFTP).
2. **Patch Management:** Establish a strict, routine schedule for updating underlying operating system packages and application binaries.
3. **Network Segmentation & Firewalls:** Implement strict firewall rules (iptables) to restrict inbound traffic to only explicitly required ports, isolating sensitive hosts from general network segments.

4. **Continuous Log Monitoring:** Deploy centralized logging and monitoring agents to detect anomalous spikes, failed authentication loops, and unauthorized service execution in real-time.

6. Conclusion

This capstone project successfully demonstrated practical offensive security methodologies paired with defensive incident response capabilities. Through controlled lab execution against 192.168.56.103, key security competencies—from initial reconnaissance to threat eradication and system hardening—were successfully validated.